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AI for Bharat Hackathon

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Team Name : WeAreGroot

Team Leader Name : Krisha Rawal

Problem Statement : AI for Communities, Access & Public Impact

Brief about the idea:

The Problem

The Information Gap: India's 450 million+ informal workers (artisans, laborers, maids) operate in an economic blind spot. They lack access to fair price benchmarks and wage data, creating severe information asymmetry

Exploitation & Undervaluation: Due to this gap, middlemen leverage their knowledge to underpay workers. For example, artisans often sell sarees worth ₹2,500 for ₹800 because they lack the means to discover the true market value

Missed Financial Support: Workers miss out on critical government safety nets (like PM-KISAN or Weaver Welfare schemes) worth ₹6,000 to ₹2.5 Lakh annually simply because they are unaware of their eligibility or cannot navigate complex application processes

The Solution

Pragati Connect is an AI-powered platform/app designed to bridge the gap between informal workers and the formal economy. It uses a "Central Brain" architecture to deliver high-tech intelligence through low-tech, accessible interfaces

A toll-free audio interface that allows low-literacy users to query local wage rates, get informed about government schemes and practice negotiations with an AI simulator. It builds confidence and ensures users know their worth before entering a deal.

A Proactive WhatsApp assistant that automatically matches users to government schemes based on their trade and location. It pushes alerts for benefits and answers eligibility questions in real-time

1.How is it different from existing ideas?

Most existing platforms are either government information portals or business directories — and they are often complex, text-heavy, and urban-focused.
This solution stands out because it:
1) Is **AI-first, voice-first, and conversational**
2) Combines government access + local business empowerment in one ecosystem
3) Works smoothly on low bandwidth and basic smartphones
4) Integrates AI calling + WhatsApp automation
It is not a website people have to understand — it understands them.

2.How will it solve the problem?

The real problem is information gap + digital literacy gap.
This app solves it by:
1) Allowing users to ask questions in their own language
2) Giving clear, step-by-step scheme guidance
3) Helping small vendors create digital presence effortlessly
4) Offering a voice-call AI assistant for non-typers
5) Using WhatsApp, a platform people already trust
It removes complexity and reduces dependency on middlemen.

3.USP of the Solution

An ultra-lightweight, multilingual AI access platform for grassroots India.
Core strengths:
1) Voice + Chat + Call + WhatsApp in one place
2) Hyperlocal and district-level relevance
3) Built for low data usage
4) Designed for low digital literacy
5) Bridges governance access and economic upliftment
It functions as a digital infrastructure layer for community empowerment; not just an app.

1. **AI Government Scheme Assistant**
Voice/text-based queries in local languages with simple eligibility guidance and step-by-step process explanation.

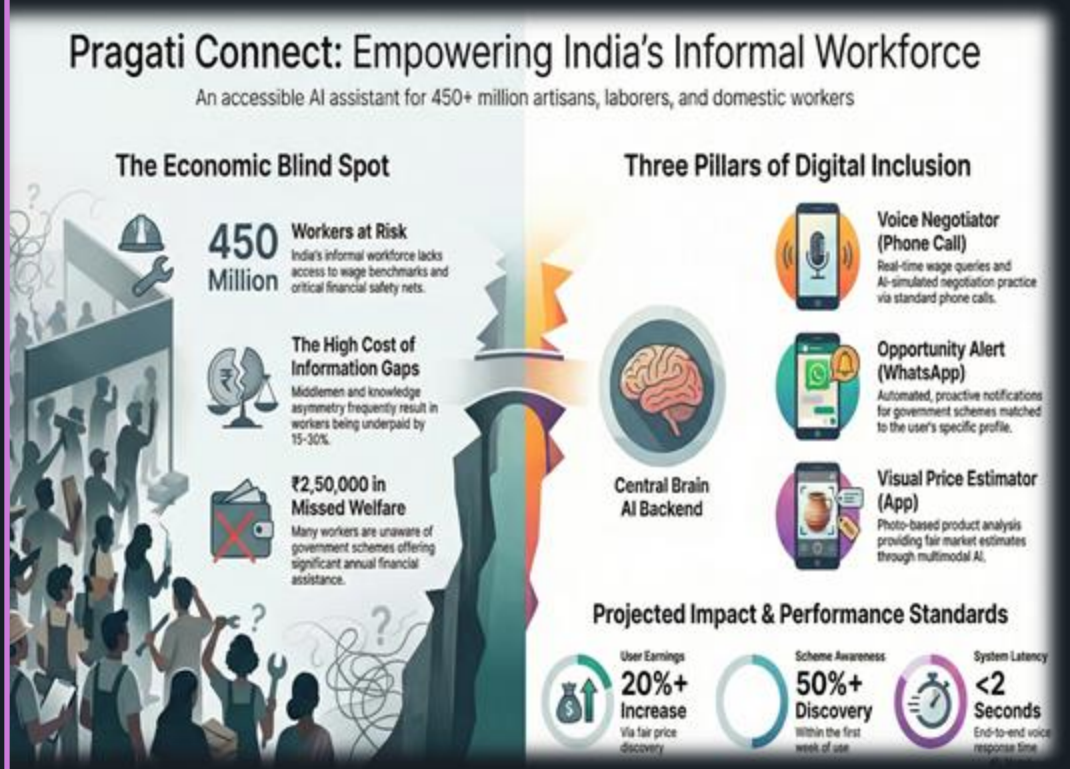
2. **AI Voice Call Support**
One-tap call feature for users who cannot read or type; supports regional languages and low-bandwidth usage.

3. **WhatsApp AI Integration**
Instant scheme and business support directly through WhatsApp — no heavy app usage required.

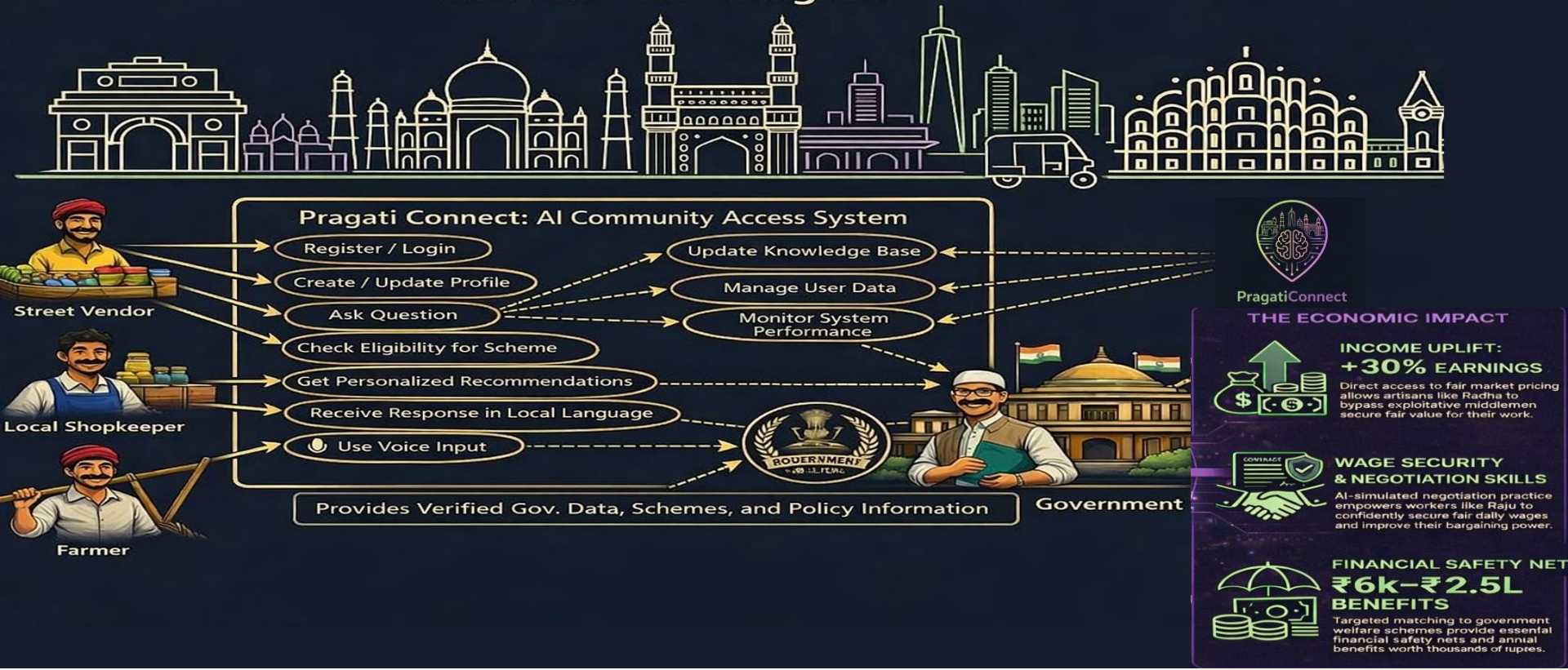
4. **Local Business Boost Platform**
Micro business profile creation, AI-generated descriptions, and geo-based visibility.

5. **Hyperlocal & Multilingual Access**
District-level filtering, community updates, and support in regional languages.

6. **Ultra-Lightweight Design**
Optimized for low-end devices with minimal data consumption.



Pragati Connect:
AI-powered Community Access Assistant
UML Use Case Diagram



Mock diagrams of the proposed solution:

App Interface

WhatsApp Assistant

On-Call Assistance

Architecture diagram of the proposed solution:

USER INTERFACES
(THE ACCESS LAYER)

Voice Negotiator via Vapi.ai

Toll-free phone interface for low-literacy users to practice negotiations and query wages with <2s end-to-end latency.

WhatsApp Opportunity Alert

Chatbot on existing number for procctive government scheme notifications and Q&A.

Visual Price Estimator

Mobile app enhancement to upload product photos for multimodal AI analysis of market value.

INTEGRATION LAYER
(THE CENTRAL BRAIN)

AWS API Gateway

Entry point handling RESTful endpoints, request validation, and webhook signatures.

AWS Lambda (Central Brain)

Serverless Python/FastAPI logic for intent classification, user profile management, and intelligence coordination.

INTELLIGENCE & DATA LAYER
(THE KNOWLEDGE CORE)

Amazon Bedrock
(Claude 3.5 Sonnet)

Reasoning engine for high-accuracy vision analysis and natural language processing.

Knowledge Base
(RAG - Gov Schemes)

Retrieval-Augmented Generation performs semattic searches across government policy documents.

DynamoDB
(User Profiles & History)

Serverless database for user trades, location, language preferences, and single digital latency.

Technologies Used:

AI & Intelligence

- Amazon Bedrock
- Claude Sonnet 3.5
- Code Gemma (Open Source)

Cloud Infrastructure

- AWS Lambda
- AWS API Gateway
- AWS Dynamo DB
- AWS S3

Voice & Communication

- WhatsApp Business
- Eleven Labs (Text to Speech)
- Deepgram (Speech to Text)
- Vapi AI (Voice Pipeline

Frontend & Backend

- Python
- React Native
- Fast API

Estimated Complete Production-Ready Implementation Cost (Monthly):

1. Voice Intelligence Stack:	
- Vapi.ai	₹830
- Eleven Labs	₹450
- Deepgram	₹72
- Rental Phone Number	₹160
2. Communication Layer:	
- WhatsApp Business API	₹800
- AWS API Gateway	₹5
3. Central Brain & AI:	
- Amazon Bedrock	₹500
- AWS Lambda	Initially Free
- Dynamo DB	Initially Free

Total + Buffer (10%) = ₹5299

Grassroot adoption & Trust model :

Grassroot Adoption

-- Partner with Panchayats & local NGOs for community onboarding
-- Train village youth as Digital Ambassadors
-- Enable WhatsApp & call-based access (no tech barrier)
-- Support regional languages for inclusive reach
-- Conduct village demo camps to build awareness & trust

Trust Model

-- Information sourced only from official government portals
-- Verified badges for registered local businesses
-- Transparent AI responses with confidence indicators
-- Simple privacy policies explained in regional language
-- Option for human assistance when required

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Thank You